SGDAgent Reasoning Loop

Agent (Step 1):
Calling tool **Global-Level Retrieval** with query "Why does the man follow the woman in red after they meet for the third time?"

Global-Level Retrieval:
Matched Events:
evt_0055 [23:45–24:40] score=2.93, confidence=0.84
man, building, woman; actions: follow; support: 2 segment(s)
evt_0025 [10:50–11:20] score=0.36, confidence=0.76
fairy, flute; actions: holding; support: 1 segment(s)
……

Agent (Step 2):
Calling tool **Segment-Level Retrieval** with query "Why does the man follow the woman in red after they meet for the third time?"

Segment-Level Retrieval :
Matched Segments:
seg_0058 [23:45–24:15] score=2.61
entities: woman, shoe, building, man; predicates: in_front_of
seg_0020 [07:55–08:25] score=0.38
entities: person, tree, ledge, water; predicates: sitting_on
……

Agent (Step 3):
Calling tool **Frame-Level Retrieval** with query "woman book"

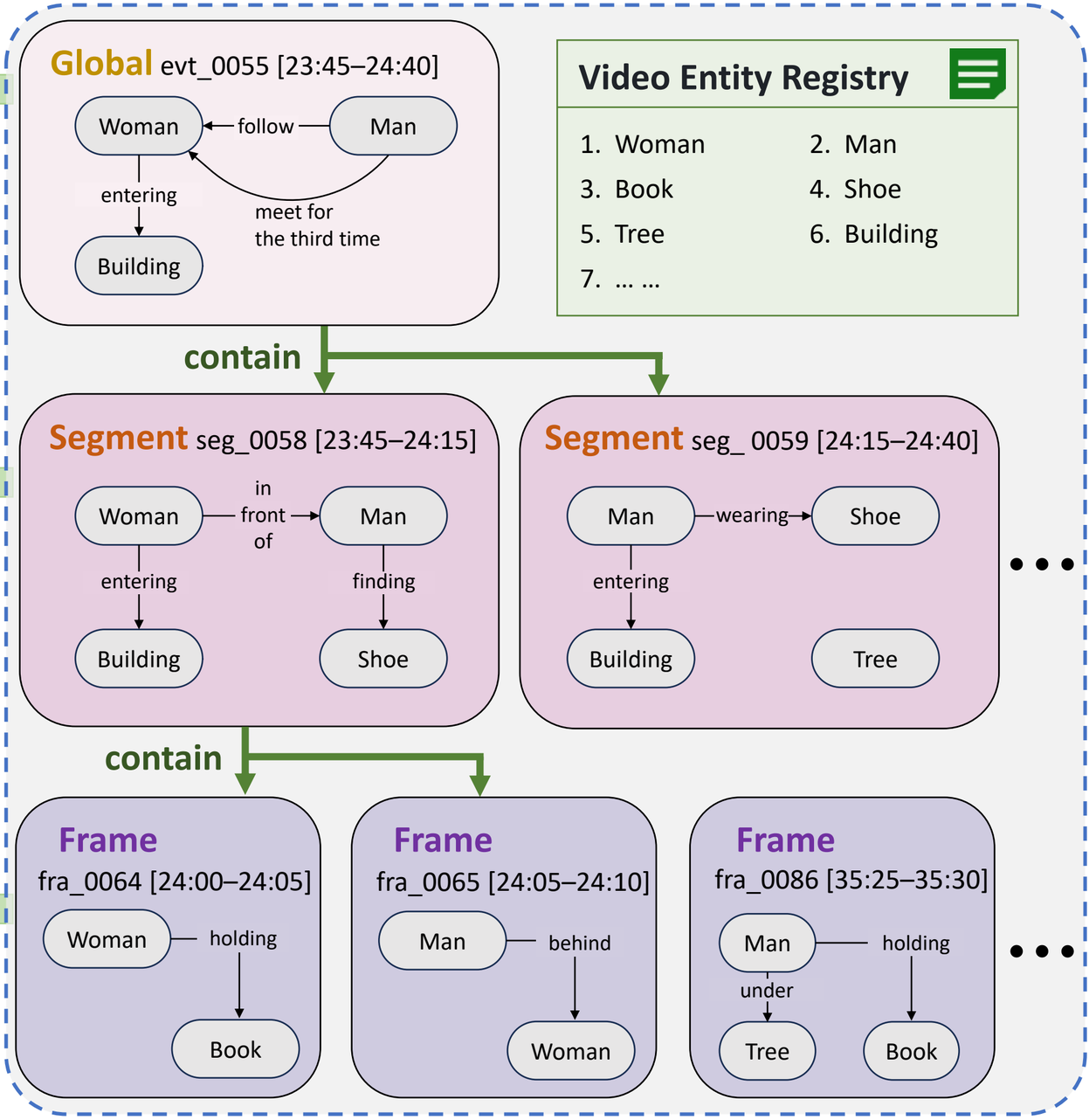
Frame-Level Retrieval :
Matched Frames:
fra_0086 [35:25–35:30] score=3.62, confidence=0.98
entities: man, book, tree; predicates: holding; Relation: (man, holding, book)
fra_0064 [24:00–24:05] score=3.62, confidence=0.96
entities: woman, book; predicates: holding; Relation: (woman, holding, book)
……

Agent (Step 4):
Calling tool **Frame Inspect** at ["00:24:00", "00:24:15"]:
"Why does the man follow the woman in red after they meet for the third time? "
"What object does she have that belongs to him?"

Frame Inspect:
Fine-grained Response:
The man follows the woman in red after they meet for the third time because she has a book that belongs to him. This indicates that he is likely trying to retrieve it or is concerned about its whereabouts, which motivates him to follow her.

Agent (Step 5):
Executing **Answer**: (D) The woman has his book.

Question: Why does the man follow the woman in red after they meet for the third time?
(A) He falls in love with the woman.
(B) He is a spy.
(C) The woman has his phone.
(D) The woman has his book.



Hierarchical Spatio-Temporal Scene Graph Representation



Long Video Input